



*Better study,
better decisions.*

Sadahalli Site Report

Site 4 · Sy. 168–190 · 76.4 acres

SITE ANALYSIS

Prepared for Qnit case-study portfolio
Presented by GeoAce Studio Pvt Ltd

Sadahalli, Devanahalli taluk, Bengaluru Rural · 13.199961° N, 77.649872° E
10 Oct 2026 / Rev A / QNIT-BLR-S04





The site at a glance

Site 4 — Sadahalli / Sadahalli, Devanahalli taluk, Bengaluru Rural



LATITUDE

13.199961° N

LONGITUDE

77.649872° E

STUDY EXTENT

1,251 m

North up · scale bar on image
Imagery Wayback 24 Sep 2026

SOURCE / Esri World Imagery (Wayback 2026-09-24). Boundary from Case_Studies.kml, Site 4; vertices V1–V18, centroid in WGS 84.

SITE AREA

309,050 m²

76.37 acres · 30.90 ha · 3,055 guntas

PERIMETER

3,232 m

KML boundary, Oct 2026

BOUNDARY

18 edges

Irregular polygon, about 883.0 × 742.0 m

Design brief

A 30.9 ha flat, open parcel on NH-44 at Sadahalli Gate, 3.9 km west of the airport's north runway. How much can it carry under the airport's height surfaces, and what do the zoning, water and boundary gaps mean for a large mixed-use scheme?

Primary opportunity

Direct frontage on NH-44 and the airport trumpet (25 m). The planned Doddajala metro station is about 1.4 km away, and the Aerospace Park, Business Park and ITIR are nearby: a strong address for hospitality, office or airport-linked mixed use.

Key constraint

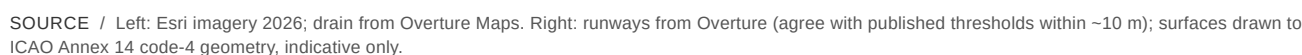
Inside the airport's 4 km inner horizontal surface: an indicative top of about 960 m AMSL leaves 32–56 m above ground. The south edge lies in the 09L approach funnel. AAI NOC is mandatory.

REPORT ROUTE

03–04 Site & ground 05–08 Environment
09–11 Context & services 12 Priorities & sources

Stakeholders: Qnit case-study portfolio / GeoAce Studio design team / Reviewer: pending

Site 4 — Sadahalli / Sadahalli, Devanahalli taluk, Bengaluru Rural



Development controls

Permitted use / source **BIAAPA RMP 2021 · zone not online**

FAR / coverage **Per zone · zonal certificate needed**

Change of land use **BIAAPA** → **BMRDA** → **state**

Height limit / basis ≈ 960 m AMSL · AAI NOC

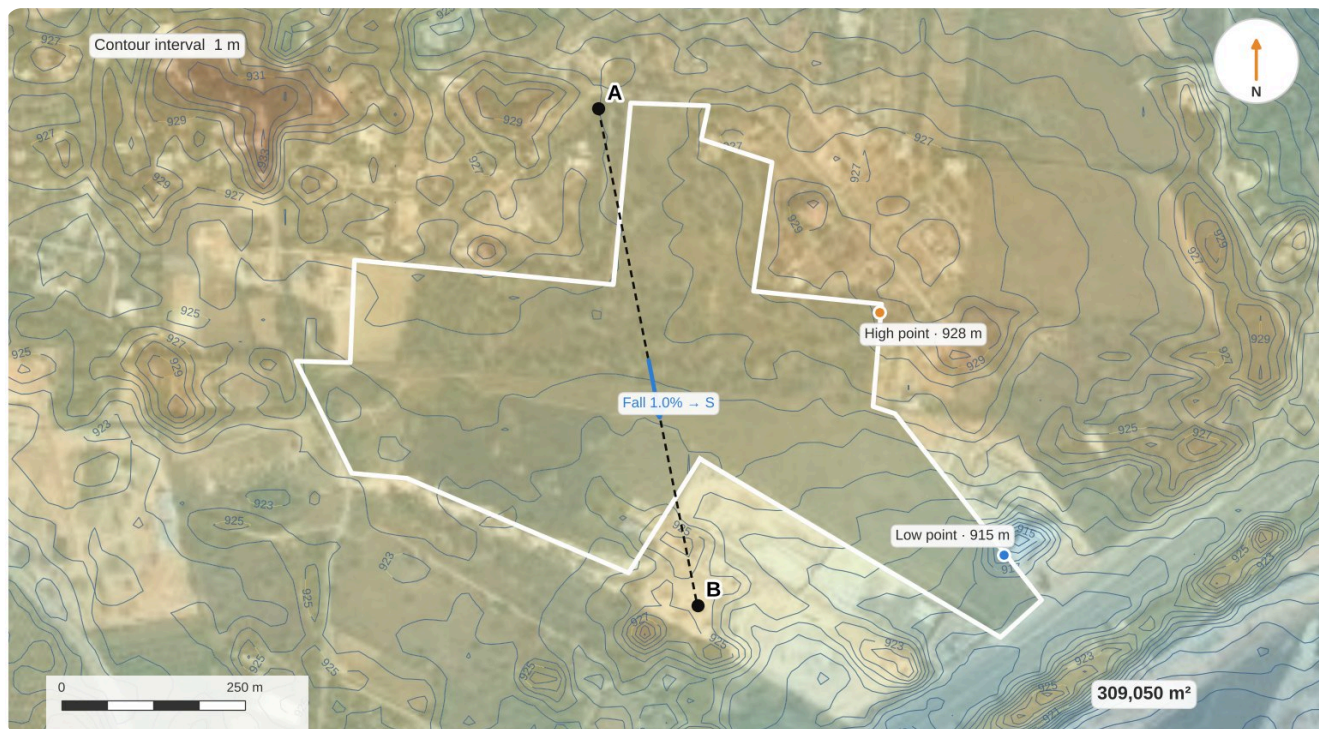
Edges	Adjoining condition	Clearance / width	Evidence
R–C · S / SE (738 m)	Service road; construction sites (Concorde, Brigade HRC Atrium, Marriott); NH-44 frontage at B–C	NH-44 25 m; drain 44 m S; metro depot 138 m	Overture land use · Esri 2026
C–K · E / N (1,150 m)	Gated villa plots on residential lanes	Lanes on H–K; villa layout overlaps 5,862 m²	Overture · Esri 2026
K–O · NW (888 m)	Track, scrub, a few houses; orchard to the W	Orchard 19 m W	Overture land use
O–R · SW (575 m)	Sadahalli Main Road (O–P); scrub; Aparna (u/c) beyond	Aparna 17 m W	Overture · Esri 2026

Height and zoning, not edges, set the envelope. Get the AAI NOCAS colour-coded zoning value and the BIAAPA zonal certificate first. Then check the boundary: Overture maps 5,862 m² of a villa layout and 2,208 m² of a neighbouring project inside the KML.



Terrain & the ground beneath

Site 4 — Sadahalli / Sadahalli, Devanahalli taluk, Bengaluru Rural



SOURCE / Copernicus GLO-30 DSM (30 m, resampled to 10 m), 1 m contours, over Esri imagery. A–B follows the fitted fall line.

ELEVATION RANGE

915 / 928 m

EGM2008 · DSM includes canopy

TYPICAL SLOPE

1.0 %

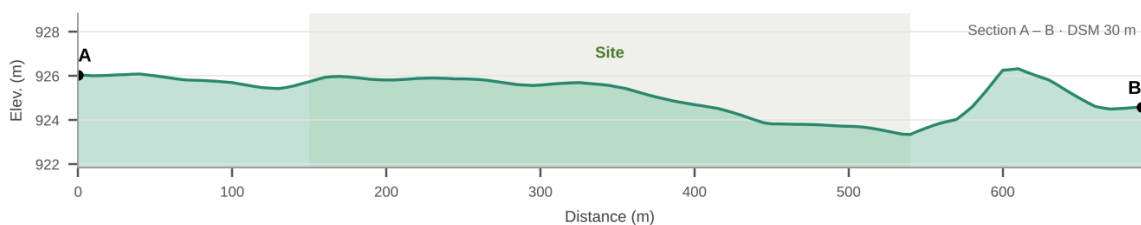
Flat · falls to the S

CONTOUR INTERVAL

1 m

GLO-30 DSM / 30 m

Transect A–B



SOURCE / Section A–B sampled every 5 m from the DSM; A is north, B south.

Section length 691 m / Relief 3.0 m / Vertical exaggeration $\approx 11\times$

DESIGN / DEVELOPMENT IMPLICATION

Essentially flat: 97% of the ground is under 5%, falling about 1% to the S, with the low point (915 m) at the SE corner by NH-44. Large floor plates need little grading. Because the ground is near 925 m, every metre of fill eats into the airport height limit.



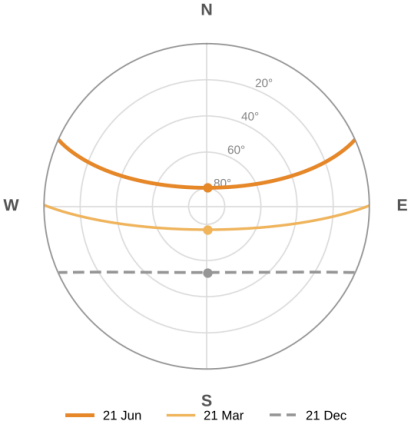
Light, shade & orientation

Site 4 — Sadahalli / Sadahalli, Devanahalli taluk, Bengaluru Rural

01 / SUN PATH



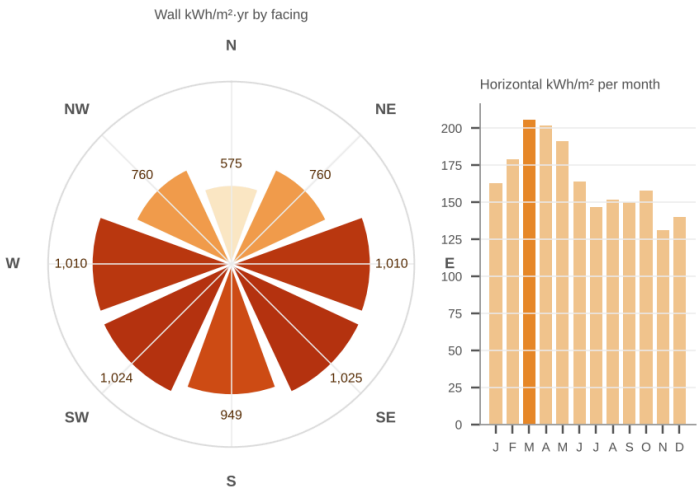
SEASONAL COMPARISON



21 Mar · 21 Jun · 21 Dec; IST (UTC+5:30)
True north; noon altitude 53° (Dec) to 80° (Jun, sun to the north)
Day length 11.3–12.9 h; no context model

SOURCE / Solar positions computed (NREL SPA, pvlib) for the site centroid; stereographic dome, Esri imagery. Hour lines at the 21st of each month.

02 / SOLAR RADIATION



ANALYSIS PERIOD

2015–2024 monthly means

INCIDENT SOLAR ENERGY

1,983 kWh/m²

Horizontal plane, per year (5.43 kWh/m²·day)
Highest wall exposure: SE / SW ≈ 1,025; north 575
Best month Mar; cloud cuts 17% vs clear sky

SOURCE / NASA POWER CERES SYN1deg monthly GHI 2015–2024; wall split modelled from clear-sky geometry scaled to it.

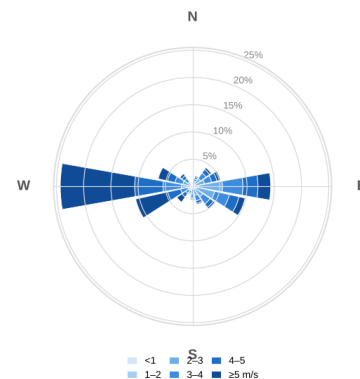
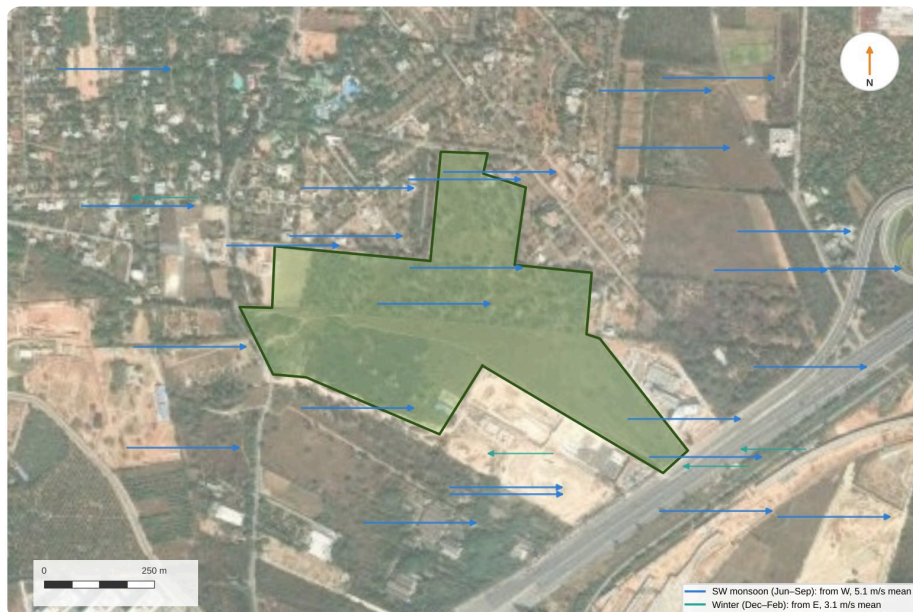
DESIGN / DEVELOPMENT IMPLICATION

With no slope or shade to work around, orient long blocks E–W with main glazing N and S: the noon sun stays above 53°, so short overhangs shade both. The 31 ha of roofs and parking at 1,983 kWh/m²·yr suit large-scale PV, subject to an AAI glare check.



Air movement & outdoor comfort

Site 4 — Sadahalli / Sadahalli, Devanahalli taluk, Bengaluru Rural



SEASON / HEIGHT

2015–24 / 10 m

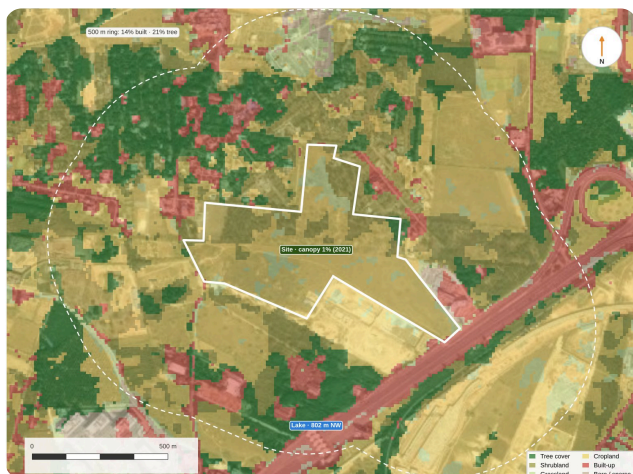
Annual: from W, mean 3.9 m/s, calm 2.8%

Jun–Sep: W 51%, 5.1 m/s

Dec–Feb: E 34%, 3.1 m/s

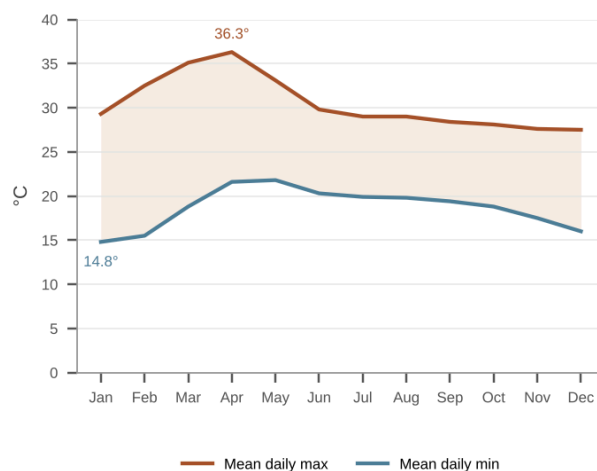
SOURCE / NOAA ISD hourly observations, Kempegowda Intl Airport (VOBL), 5.4 km E; consistent with NASA POWER MERRA-2 (from W, 3.9 m/s).

Local comfort



SOURCE / ESA WorldCover 2021 (10 m) on Esri imagery; dashed = 500 m ring.

Seasonal temperature



SOURCE / NASA POWER MERRA-2 daily, 2015–2024. VOBL station mean 24.2 °C vs model 24.0 °C.

DESIGN / DEVELOPMENT IMPLICATION

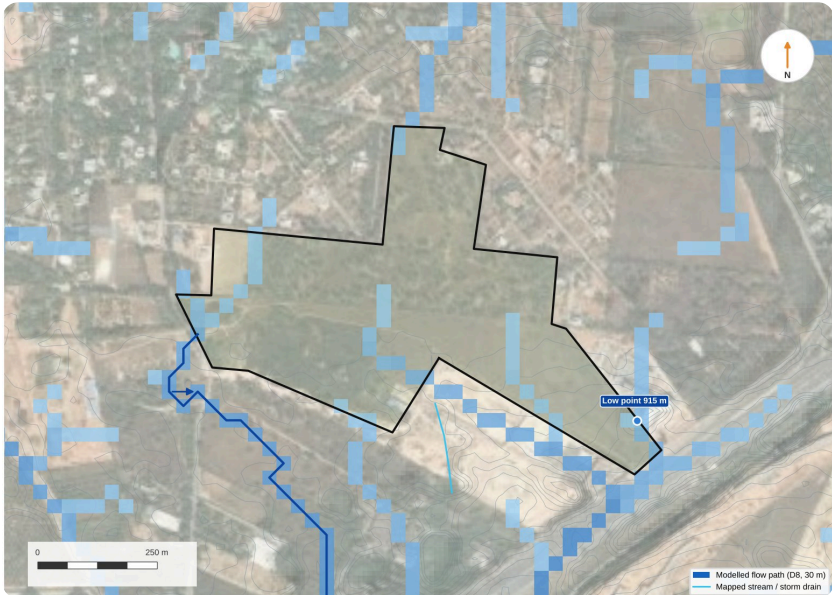
The site is open to the monsoon westerly (5.1 m/s, 51% of Jun–Sep hours): use E–W courts to ventilate it. Winter easterlies carry NH-44 traffic noise and dust, so put a planted buffer and parking along the SE frontage. Canopy is only 1%: plan shade trees early.



Runoff, storage & groundwater

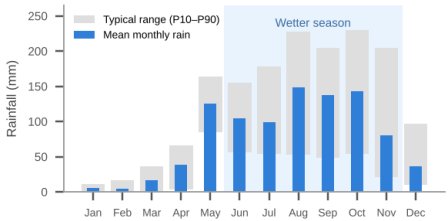
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SURFACE RESPONSE



SOURCE / D8 flow on GLO-30 DSM; Overture water. Rain: NASA POWER MERRA-2 grid 13.0 N 77.5 E (taluk normal 804 mm, CGWB).

MONTHLY RAINFALL

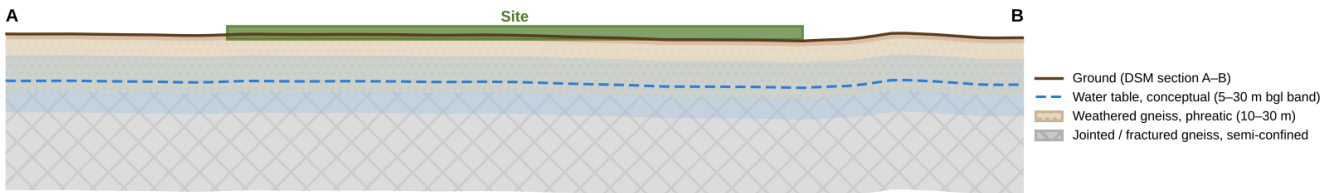


EVENT BASIS

95 mm / 24 h

10-year return period (Gumbel on 1981–2025 annual maxima)
Mean annual 937 mm; 65% falls May–Nov

Groundwater & recharge



SOURCE / Conceptual section on A–B. Strata and water depths from the CGWB Bengaluru Urban profile, not site boreholes.

Surface water

Low point SE corner (915 m) by NH-44 → mapped drain 44 m S.
10-yr day on 30.90 ha: $\approx 10,300 \text{ m}^3$ runoff now (C 0.35), $\approx 22,000 \text{ m}^3$ built (C 0.75).
Basis: catchment = site; 13 ha drains through the outlet.

Ground conditions

Devanahalli extraction 169% of recharge (2024): over-exploited.
Shallow aquifer largely desaturated (CGWB NAQUIM 2022); levels 20–60 m bgl.
Granite-gneiss; no Cauvery supply in the taluk.

DESIGN / DEVELOPMENT IMPLICATION

Hold $\approx 11,700 \text{ m}^3$ of extra 10-year runoff in a detention park at the SE low point before the NH-44 drain. Water is the delivery risk: plan on treated-water reuse, HN Valley-fed tanks and harvesting, not borewells.



Living systems & views

Site 4 — Sadahalli / Sadahalli, Devanahalli taluk, Bengaluru Rural



SOURCE / Esri imagery; Overture water bodies; plan-view sight lines from the site centre.

Landscape structure



Landform



Vegetation



Built form

Canopy cover **1% site · 21% 500 m ring**
Permeable ground **100% site · 86% ring**
Water / habitat edge **drain 44 m · tanks 802 m NW**
Tree survey / season **none · WorldCover 2021**

View & habitat priorities

Retain

The scattered trees and scrub belt across the centre, as the seed of a central park.

Investigate

Bird activity near the runways (wildlife-strike rules for open water and refuse) and aircraft noise; no published noise contour was found.

Observer: site centre, plan view (no eye-height model)
WorldCover 2021: 66% cropland, 22% shrub

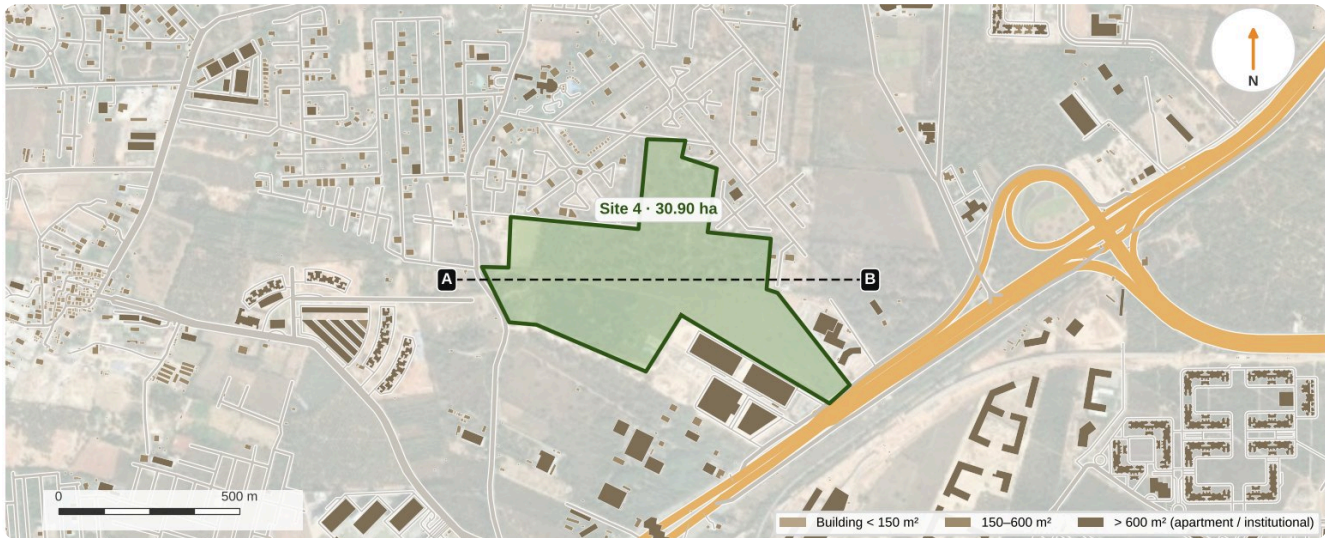
DESIGN / DEVELOPMENT IMPLICATION

Make a shaded central green from the existing tree belt, and keep detention basins dry-bottomed rather than open water, which attracts birds this close to the runways.



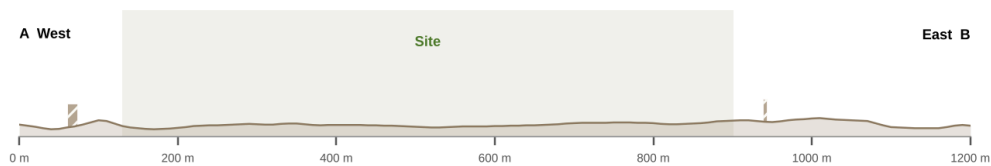
Neighbourhood fit & movement

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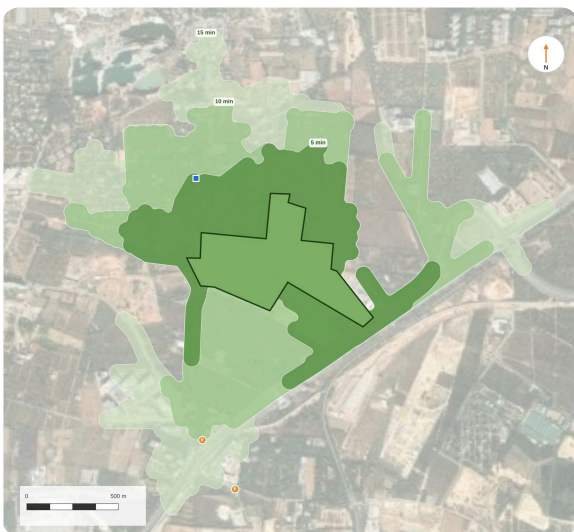
SOURCE / Overture Maps buildings and roads (release 2026-09-23) over Esri imagery; A-B is a 1.2 km W-E section through the site.

STREET & SKYLINE PROFILE A-B



2 buildings cut; heights are tagged for only 5% of buildings within 500 m (up to 54 m), so hatched blocks are assumed from footprint.

DAILY ACCESS



Walking 5 / 10 / 15 min at 4.8 km/h on the Overture road network from the nearest road nodes.

ACCESS BASIS

NH-44 frontage / walk / weekday

Transit **Jade Garden stop** · 299 m

Rail **KIA Halt** · 2.8 km

Metro **Doddajala (Blue 2B)** ≈ 1.4 km, 2027-28

Health **Trust Aid Hospitals** · 26 min

Education **Akash International School** · 11 min

Daily needs **none within 1 km mapped**

NH-44 is tolled at Sadahalli and a ₹35 crore underpass is planned at the junction: confirm the legal access point and whether the frontage gets a direct entry.

Context: villas to the N, hotels and offices rising to the S and E.

Overture 2026-09-23.0 · checked 10 Oct 2026



The context beyond the boundary

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LAND-USE CHANGE

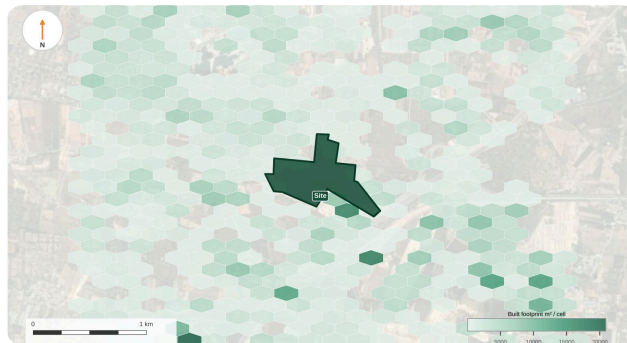


SOURCE / Esri Wayback releases 27 Feb 2017 (left) and 24 Sep 2026 (right).

Recorded change

2017 → 2026: the parcel stays open land; the villa layout to the N filled in and large projects started on the S and E.
WorldCover 2021: 66% cropland.
Heritage: Devanahalli Fort (c.1501, Tipu Sultan's birthplace) in the taluk; none on site.

PEOPLE & ACTIVITY



SOURCE / Built-footprint density from Overture buildings, 2026; activity proxy.

Community needs

Census 2011 figures for Sadahalli conflict (2,511 vs 47); Devanahalli taluk 209,622.
Daytime: airport, Aerospace Park and Business Park workforces.
No shops or clinics within 1 km mapped: retail and health fit the programme.

Local architectural vocabulary



Front yard

Tiled house, Puttur



Veranda

Pillared front, Melukote



Roof form

Clay-tile roofs, Krishnarajapet



Street edge

Jagali steps, Malavalli

Shaded verandas and courtyards keep the region's buildings cool under a hot, open sky: a cue for low, deep-plan buildings under the airport ceiling.

Wikimedia Commons, CC BY-SA 4.0: Prof tpms (Melukote, Krishnarajapet, Malavalli); Bharathesha Alasandemajalu (Puttur) / Old Mysore and coastal Karnataka examples, not on-site records

DESIGN / DEVELOPMENT IMPLICATION

Under a ~35 m ceiling, favour courtyard blocks with shaded colonnades along the internal streets, rather than towers.



Networks, connections & gaps

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SOURCE / Overture Maps infrastructure and roads (OpenStreetMap-derived) over Esri imagery. Network capacity needs provider records.

Service register

Service	Connection / distance	Capacity / status	Evidence / next check
Water	No piped main mapped; no Cauvery in taluk	Aquifer over-exploited (169%)	KIADB / BIAAPA: bulk or treated supply
Power	Substation 683 m; HT line to BIAL 991 m	Load not confirmed	BESCOM: feasibility letter
Drainage	Outfall SE corner → drain 44 m S (NH-44 side)	Capacity unknown	NHAI / GP: outfall consent
Sewer / waste	No sewer mapped	On-site STP needed	KSPCB consent; GP collection

DESIGN / DEVELOPMENT IMPLICATION

Power is close; water and sewer are not. Size an on-site STP with treated-water reuse for flushing and landscape, and secure bulk water before phasing occupancy.



What matters for this project

Site 4 — Sadahalli / Sadahalli, Devanahalli taluk, Bengaluru Rural

01

Airport height ceiling

Evidence p03 Inside the 4 km inner horizontal ($\approx$ 960 m AMSL): 32–56 m above ground, lowest at V6; the south edge sits in the 09L approach funnel. Indicative only.

Next: AAI NOCAS colour-coded zoning value for each corner Owner: architect Due: concept stage

02

Zoning and boundary

Evidence p03 The BIAAPA RMP 2021 zone for Sy 168–190 is not online. Mapped neighbouring projects overlap the KML by 8,070 m²; Sy 190 is 7 m outside.

Next: BIAAPA zonal certificate; survey sketch vs KML Owner: client surveyor Due: before purchase

03

Water and sewer

Evidence p07, p11 Devanahalli extraction 169% with no Cauvery supply; no sewer mapped.

Next: bulk-water source and STP strategy Owner: services consultant Due: feasibility stage

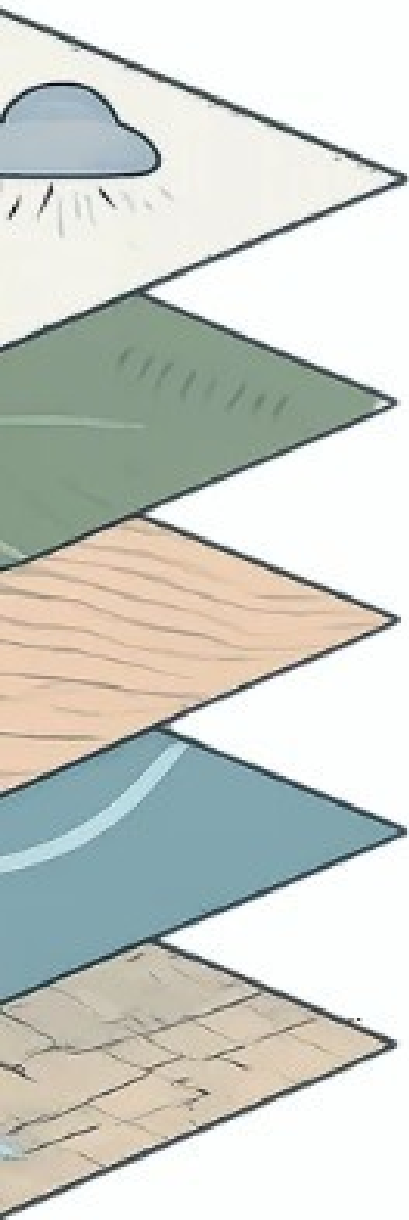
Evidence & coverage

Analysis	Dataset / date	Method / resolution	Status / limitation
Site / controls	KML Oct 2026; RCCMS; research on BIAAPA and airport rules	UTM 43N; ICAO Annex 14 surfaces	Zone not found; heights indicative
Terrain / water	Copernicus GLO-30; Overture water	30 m DSM; D8; Gumbel 1981–2025	DSM includes trees; no site survey
Climate / solar	NASA POWER MERRA-2, CERES 2015–24; NOAA ISD VOBL	0.5° / 1° grid; station 5.4 km	Regional, not microclimate
Context / services	Overture 2026-09; Esri Wayback 2017 / 2026	Walk 4.8 km/h network; 1.2 km section	Few heights; capacity unconfirmed

Not run / unavailable: BIAAPA zoning map and regulations (host blocked); AAI CCZM grid value; soil texture (SoilGrids returned 503); noise contours; site borehole and topographic survey.

Prepared by Qnit analysis · GeoAce Studio Reviewed by: pending

Analysis run S04 / 2026-10-10 04:08 UTC Boundary version Case_Studies.kml (Oct 2026)



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starts
with
you.**

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